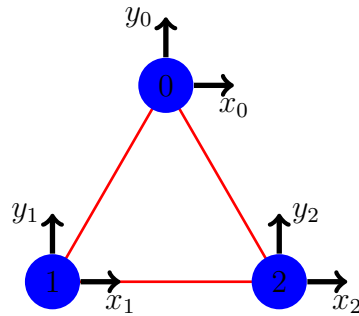
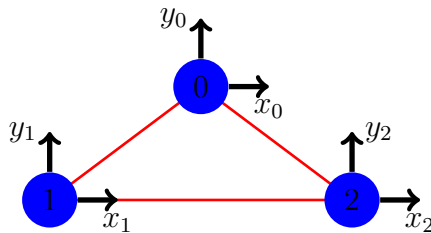


Homework 6

- Let us keep working on the triangular molecule with in-plane displacements, building on what was done in the last homework.



- Write the stiffness matrix in the basis of irreducible representations.
 - Write the potential energy in terms of the irreducible derivatives and the symmetrized displacement amplitudes (i.e. slide 77 from group theory notes).
- Assume that the triangle was compressed along the vertical axis and the symmetry was broken.



- Find all of the group elements that leave the above triangle invariant (ie. point group).
 - Construct a representation of the group for the in-plane displacements.
 - Decompose this representation into irreducible representations. Which ones are present? Will there be any degeneracies?
 - For each irreducible representation vector, construct the symmetry lineage to find the parent within C_{3v} .
 - Write out the expression for the potential energy as you did in part 1b.
- This problem deals with Graphene (see schematic).

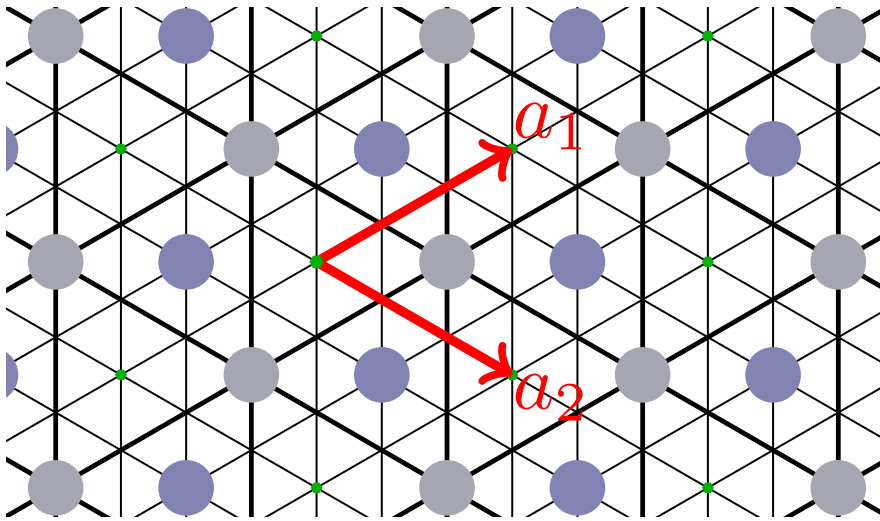


Figure 1: Crystal structure of graphene. The large dots are carbon atoms.

- Assume that the nearest neighbor distance between like colored atoms is a . Write out the vector equations for the two pictured lattice vectors.
- Define the basis atoms in direct coordinates.
- Construct the reciprocal lattice vectors for graphene. Make a sketch of the First Brillouin zone (i.e. the Wigner-Seitz unit cell for the reciprocal lattice).
- Assuming that $N_1 = N_2 = 3$, plot all allowed k-points within the first Brillouin zone.